

Established findings the theory builds on

Word frequency and predictability are benchmark predictors of reading times, with similar effects across languages

(Kuperman et al., 2024; Smith & Levy, 2013)

Long syntactic dependencies tax working memory, and the load weighs most heavily on developing readers

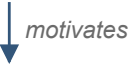
(Gibson, 1998; Delage & Frauenfelder, 2019)

Breaks in discourse coherence force bridging inferences that cost reading time

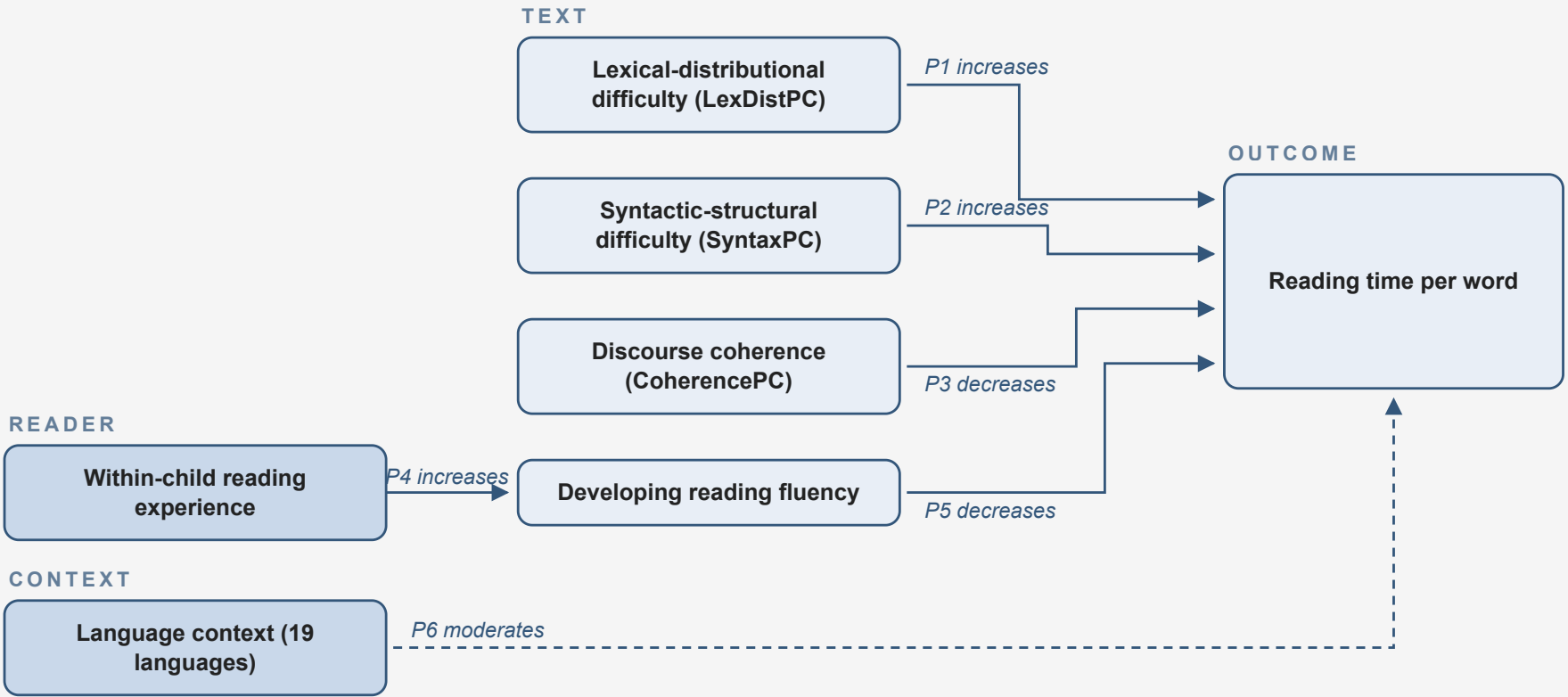
(Kintsch, 1988; McNamara et al., 1996)

Reading practice builds fluency through automatization and self-teaching

(LaBerge & Samuels, 1974; Share, 1995; Mol & Bus, 2011)



The theory: constructs and labelled propositions



Predictions, each pre-registered and tested in the confirmatory analysis

Hypothesis 1a

Stories higher in syntactic-structural difficulty are read more slowly: $\beta(\text{SyntaxPC}) > 0$, beyond the ROPE.

weighed against constant-rate pacing

Confirmatory model, HDI+ROPE

Hypothesis 1b

Stories higher in lexical-distributional difficulty are read more slowly: $\beta(\text{LexDistPC}) > 0$, beyond the ROPE.

weighed against constant-rate pacing

Confirmatory model, HDI+ROPE

Hypothesis 1c

More coherent stories are read faster: $\beta(\text{CoherencePC}) < 0$, beyond the ROPE.

weighed against constant-rate pacing

Confirmatory model, HDI+ROPE

Hypothesis 2

Reading speeds up within child as reading experience accumulates: $\beta(\text{Experience}) < 0$, beyond the ROPE.

weighed against skimming / disengagement and selective retention

Confirmatory model + engagement check

Alternative accounts weighed by the same tests:

Constant-rate pacing

Skimming / disengagement

Selective retention

Unmeasured account, bounds interpretation:

Narration pacing